Hydraulic Oil System

The hydraulic system consists of a two pump stack directly coupled to the engine. A hydraulic valve block (manifold)is provided for quick and easy testing and troubleshooting.

Hydraulic oil is filtered by a screen filter located in the tank filler neck, a ***40 micron suction filter*** located in the tank, and a ***10 micron return filter***, with cold oil bypass valve located in the return circuit.

It is recommended that ***ISO 46 type hydraulic oil*** or equivalent be used when adding or replacing the hydraulic oil is required.

DO NOT USE MULTI-VISCOSITY OIL. Cleanliness is a very important part of proper hydraulic system operation. Hydraulic oil is not only used to transfer power; it also lubricates and cools the system components. Keeping the hydraulic system clean can help reduce costly repairs.

The hydraulic oil level sight glass is located on the right rear of the front drum, below the engine compartment. This level should be checked daily. Oil must be below the top and above the bottom of the sight glass. **DO NOT OVERFILL!** Care should be taken to clean the filler cap before adding oil to the system. If hydraulic oil has to be added, the machine should be inspected for leaks.

The ***suction filter*** (Figure 26) is located inside the hydraulic tank. This filter is attached to the fitting connected to the hydraulic pump suction hose.

The ***return filter*** (Figure 26) is located at the front of the engine compartment. Replace both filters according to the Table 5.

⚠ CAUTION - HYDRAULIC SYSTEM HAZARD

DO NOT open hydraulic lines or loosen hydraulic fittings while engine is running! Hydraulic fluid under pressure can penetrate the skin, blind, cause burns or create other potentially dangerous hazards follow all safety instructions as described throughout this manual.

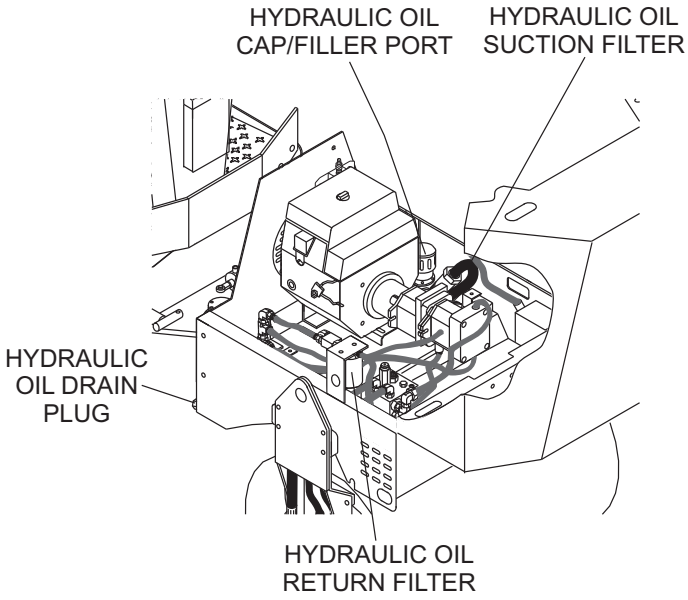


Figure 26. Hydraulic Filter Locations

Changing Hydraulic Oil and Filters

1. Park the roller on a clean flat work area and set the parking brake.
2. Remove the ***hydraulic oil drain plug*** (Figure 26) and drain the hydraulic oil. Dispose of the used oil in an environmentally friendly manner. Replace the drain plug and tighten.
3. Remove the return filter and ***install a new filter***. Dispose of the used filter in an environmentally friendly manner.
4. Disconnect the suction hose and remove the fitting from the tank. ***Replace*** the suction filter. Dispose of the used filter in an environmentally friendly manner. Replace the fitting and reconnect the suction hose.

Freewheel Engagement Valve

This hydraulic system has a freewheel engagemet valve allowing hydraulic oil to be bypassed. ***Open*** (turn allen wrench counter-clockwise) this valve (Figure 27) to engage the freewheel capability of the roller. When the valve is fully opened, ports **A** and **B** are allowed to connect, bypassing the oil to and from the drum drive motors.

Remember the freewheel engagement valve should only be used in emergencies when the roller cannot be driven due to engine or hydraulic system problems.

When towing of the roller has been completed, this valve *must* be ***closed*** (turn allen wrench fully clockwise) completely and the lock nut set. Failure to close this valve completely will result in low power, improper speed, and excessive hydraulic oil temperature.

⚠ CAUTION - Freewheel Engagement Valve

The freewheel engagement valve (towing) is only for ***emergency use***. **DO NOT** move roller over 2 MPH or long distances as hydraulic system component failure could result.

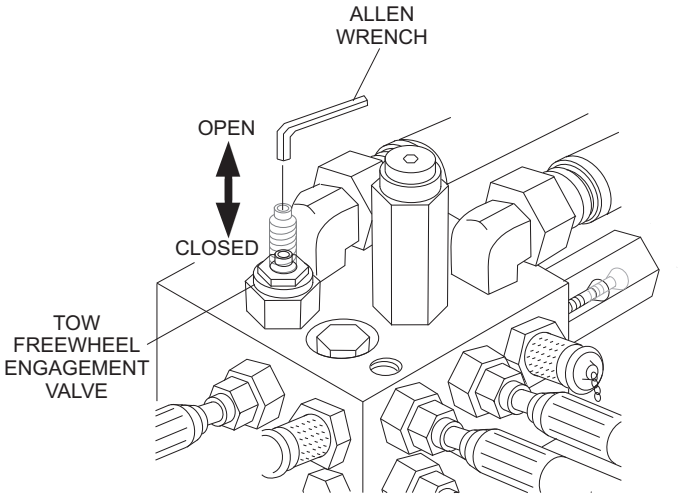


Figure 27. Freewheel Engagement Valve

Drum Drive

The drum drive circuit is a ***parallel***, ***closed loop system*** (Figures 38 and 39) consisting of a hydrostatic pump, two relief valves, a freewheel engagement valve, front and rear drum drive motors.

The hydrostatic pump is manually controlled by a cable connected to the forward/reverse shift lever located on the right side of the operator seat. When the shift lever is placed in forward, high-pressure oil is supplied by the hydrostatic pump to the valve block (***port A***). The valve block (manifold) directs this high-pressure oil to the front and rear drum drive motors. Return oil from the motors is returned to the valve block (***port B***) and is returned to the suction side of the hydrostatic pump.

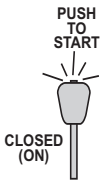
When shifted into reverse, the high-pressure and suction ports on the hydrostatic pump are reversed. Oil flow is then in the opposite direction of forward (***port B*** becomes high-pressure and ***port A*** becomes suction)

Vibration and Steering

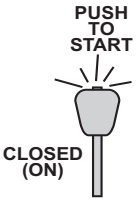
The vibration and steering system is an ***open loop circuit*** operated by a gear type pump. Separate relief valves control each circuit. This system consist of the gear pump, relief valves, electric vibration control valve, vibration drive motor, steering valve, and steering cylinder.

The vibration circuit is controlled by an electric control valve located on the valve block (manifold). This valve is controlled by the "***ON/OFF***" pushbutton switch mounted on top of the travel lever.

High-pressure oil is supplied by the pump to the valve block (***port P***) and is directed to the electric control valve. When the pushbutton switch is in the "***OFF***" position, this valve is ***open*** allowing oil to go to the steering valve, without driving the vibration motor.



When the switch is in the "***ON***" position, the electric control valve ***closes*** and oil is directed out of ***port 1*** to the vibration motor. Return oil from the motor returns to the valve block via ***port 2*** and is directed to the steering valve.



Steering is controlled by a ***steering valve*** and ***cylinder*** (Figures 29 and 30). The steering wheel is direct coupled to the steering valve controling the oil flow to the cylinder. Oil supplied from the vibration circuit is directed to ***port 3*** which connects to ***port P*** of the steering valve. When steering is not being used, oil passes out of ***port T*** of the valve block and returns to the hydraulic tank. When the steering wheel is operated, the steering valve closes and oil is directed to ports **L** or **R** to extend or retract the steering cylinder.

Rubber Scrapper Mat

Rubber scraper mats have been provided for the cleaning of the front and rear drums. Adjust the scrapers mats as close as possible to the drums, using the slotted holes (Figure 28) provided. Replace these rubber mats when they become badly worn.

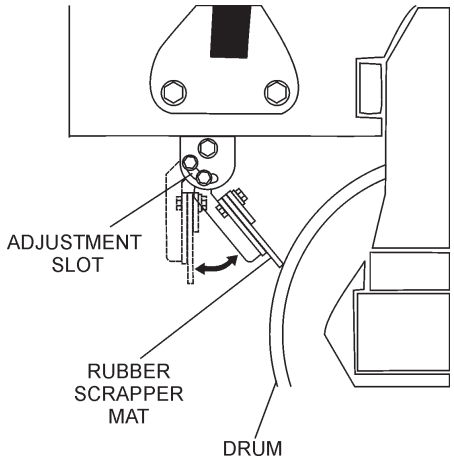


Figure 28. Scraper Bar Adjustment

Forward/Reverse Hydraulic Pressure Test

Park the machine on a solid flat surface and stop the engine. Set the parking brake and ***block the front drum securely***.

1. Check hydraulic oil level by viewing the hydraulic oil sight glass. The hydraulic oil level must be below the top and above the bottom of the sight glass. **DO NOT OVERFILL!**
2. Adjust engine RPM (3,500 +/- 50 RPM)
3. Let engine run for 5 minutes, this will bring the hydraulic oil operating temperature to a minimum of 135° farenheit.
4. Check and repair all hydraulic leaks.
5. On the manifold block install a 5,000 psi pressure gauge (Figure 34) to the ***forward*** pressure quick disconnect test port 2
6. Run engine at full throttle.
7. Move the ***travel lever*** to the forward position. **MAKE SURE FRONT AND REAR DRUMS DO NOT SPIN.**
8. Read the pressure gauge. The ***relief pressure*** (roller drums blocked) will read 2,900 ± 145 psi. Under ***normal operating*** conditions this pressure will read 400-600 psi.
9. Return the travel lever to the ***neutral*** position and ***stop*** the engine.
10. Install the pressure gauge into the reverse quick disconnect ***port 3*** and repeat the above procedures. The relief pressure reading for the **reverse** test port 3 will be the same (2,900 psi). Again normal operation pressure for the reverse port will be 400-600 psi.

    Normal operating pressures are based on the machine traveling on level, firm surface. The operating pressures will increase significantly when traveling uphill.

Vibration Circuit Pressure Test

Place the front drum on soil, gravel, or a heavy rubber mat. **DO NOT ACTIVATE VIBRATION FEATURE ON CONCRETE OR HARD SURFACE!**

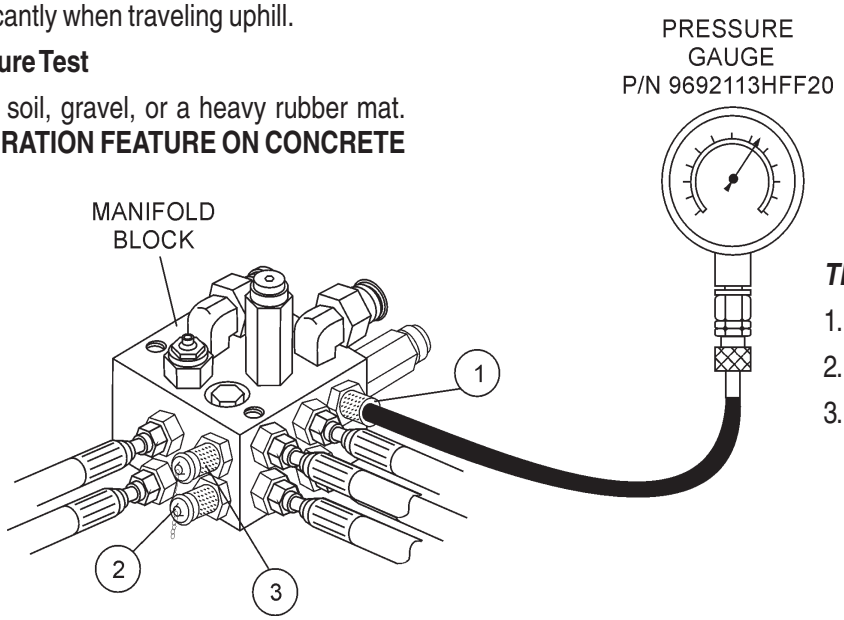
1. On the manifold block install a 5,000 psi pressure gauge to quick disconnect test port 1.
2. Start the engine and run at full throttle.
3. To start the vibration, ***press*** the pushbutton switch (vibration control) located on top of the travel lever. Under normal operating conditions this pressure will range between 900-1,500 psi.

> ***NOTICE***
>
> Vibration relief pressure is hard to read accurately using this test. It may be necessary to disconnect the pressure line to the vibration drive motor. Plug this line and re-test, pressure will be exact relief pressure. **DO NOT** perform this test for a long period of time — damage could occur.

Steering Pressure Test

1. On the manifold block, install a 5,000 psi pressure gauge into quick disconnect test ***port 1*** (Figure 34).
2. Start the engine and run at full throttle.
3. Turn the steering wheel to the ***left or right*** (maximum) and hold. Read the steering relief pressure. The ***relief pressure*** reading for the **steering** test port 1 will be 700 psi. Again, under normal operating conditions, pressure for the steering port will be between 200-400 psi.

If maximum pressure cannot be reached, plug the steering cylinder pressure hoses and re-test. If pressure is correct the steering cylinder is leaking. **DO NOT ACTIVATE THE VIBRATION FUNCTION WHILE PERFORMING THIS TEST !**



TEST PORTS

1. Vibration and Steering
2. Reverse
3. Forward

Figure 34. Manifold Test Ports

Forward/Reverse Relief Valves Drive Adjustment

The ***forward*** and ***reverse*** relief valves (Figure 35) are located in the hydrostatic pump (not on the manifold block) under plugs 1 and 2 . Plug 2 is located in the same location as plug 1, except it is on the bottom side of the pump. Relief valve 1 is ***reverse*** and 2 is ***forward***.

> ***NOTICE***
>
> Axle Pumps supplied **after 01/01/2003 DO NOT** have adjustable relief cartridges. The pressure is fixed at 2,900 ± 145 psi. If the relief pressure is not within normal range, the complete cartridge must be replaced.

1. Clean the area around the cap.
2. Remove the cap.
3. Carefully remove the valve cartridge.
4. Using an ***allen wrench***, remove the top nut (extreme care must be taken not to loosen existing shims, spring, or valve) Add or remove shims as required. ***Adding*** shims increases pressure; ***removing*** shims lowers pressure. One shim is equal to approximately 50 psi. Shim P/N is 34538.
5. Install the top nut and install valve in the pump cavity using extreme care not to bind.
6. Install the valve cap.
7. Re-test pressure, further adjustment may be needed if pressure is not correct.

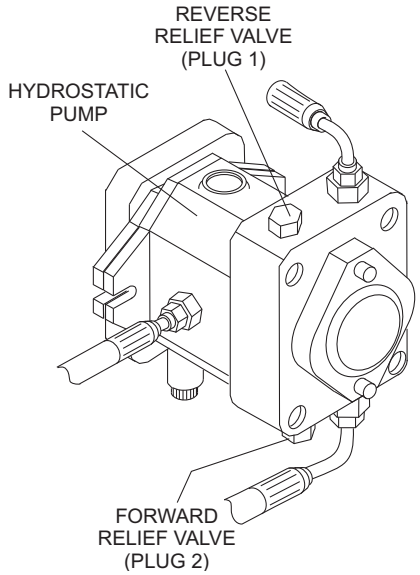


Figure 35. Adjusting Forward/Reverse Relief Valves

Steering Relief Valve Pressure Adjustment

To adjust the relief valve steering pressure perform the following:

1. Insert an 8 mm allen wrench into steering relief ***port 4*** (Figure 36) on the manifold block.
2. On the manifold, connect at 5,000 psi pressure gauge into quick disconnect test ***port 1***.
3. Start the engine and run at full throttle.
4. Turn the steering wheel to the ***left or right*** (maximum) and hold. Read the steering relief pressure. The ***relief pressure*** reading for the **steering** test port 1 will should be 700 psi.
5. If the steering relief pressure is not 700 psi, using the allen wrench adjust the pressure at ***port 4*** until the pressure gauge reads 500 psi.

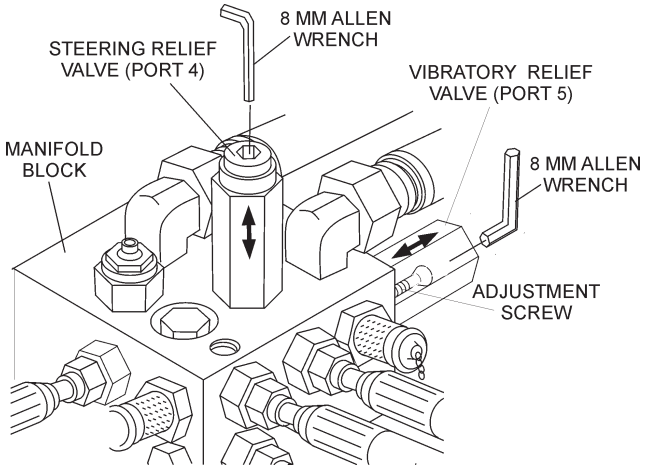


Figure 36. Steering and Vibration Relief Valves

Vibration Relief Valve Pressure Adjustment

1. Insert an 8 mm allen wrench into steering relief ***port 5*** (Figure 36) on the manifold block.
2. On the manifold block, insert a 5,000 psi pressure gauge into quick disconnect test ***port 1***.
3. Start the engine and run at full throttle.
4. To start the vibration, ***press*** the pushbutton switch (vibration control) located on top of the travel lever. The relief pressure should read 900 ~ 1,500 psi.
5. If the vibration relief pressure is not 900 ~ 1,500 psi using the allen wrench adjust the pressure at ***port 5*** until the pressure guage reads correctly.

Removing and Replacing Hydrostatic Pump

1. Set the parking brake.
2. Disconnect the battery.
3. Clean the pump and all connections.
4. Mark and disconnect all hoses and lines from the pump.
5. Disconnect the forward / reverse control cable.
6. Disconnect the pump support bracket.
7. Remove the engine mounting bolts.
8. Elevate the pump and engine assembly using a proper lifting device.
9. Disconnect and remove the hydrostatic pump assembly.
10. Repair or replace the hydrostatic pump as required.
11. Install the hydrostatic pump in the reverse order of removal, using *Locktite 271* on all mounting bolts and nuts.
12. Test operate. Test and adjust the forward and reverse relief pressures as required. Adjust the forward / reverse control cable.

Removing and Replacing Vibration/Steering Pump

1. Remove the hydrostatic pump as per preceding instructions.
2. Remove all hoses and lines.
3. Disconnect the vibration / steering pump and remove.
4. Repair or replace pump as required.
5. Install the pump in the reverse order of removal, using ***Locktite 271*** on all mounting bolts and nuts.
6. Test operate. Test and adjust the forward and reverse pressure relief valves as required. Adjust the ***forward/ reverse control cable***. Test and adjust the vibration and steering pressure relief valves as required.

Drums and Main Frame

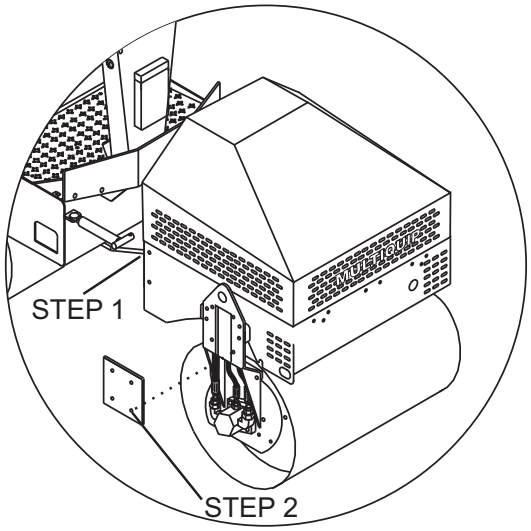
The front drum is designed to apply vibration and compaction force to the operating surface for compaction. This vibration and compaction force is produced when the vibrator shaft is rotated. Maximum efficiency is achieved only when the engine is operated at full throttle.

A single drive motor is mounted on the left side of the drum and is shockmounted. This type of drive motor is designed for maximum torque and power.

The vibrator is driven by a gear motor coupled to the vibrator shaft. The vibrator assembly rotates inside of a sealed housing containing oil to lubricate the bearings. This side of the drum is also shock mounted.

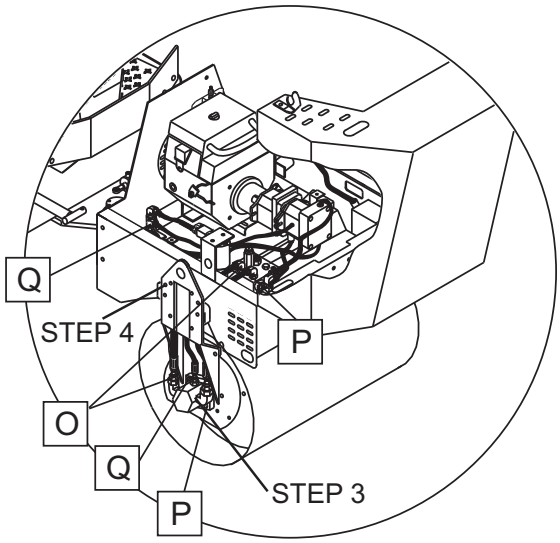
Front and Rear Drum Removal and Disassembly.

The removal and disassembly of the front and rear drums shall be referenced by Figures 37, 38, and 39 on the following pages. When reassembling the front drum use ***Locktite 271*** on all mounting hardware.



1. JACK FRAME UP SLIGHTLY AT ARTICULATION JOINT.

2. REMOVE THE FOUR ALLEN HEAD SCREWS HOLDING THE SIDE COVER OF THE DRUM SUPPORT PLATE.

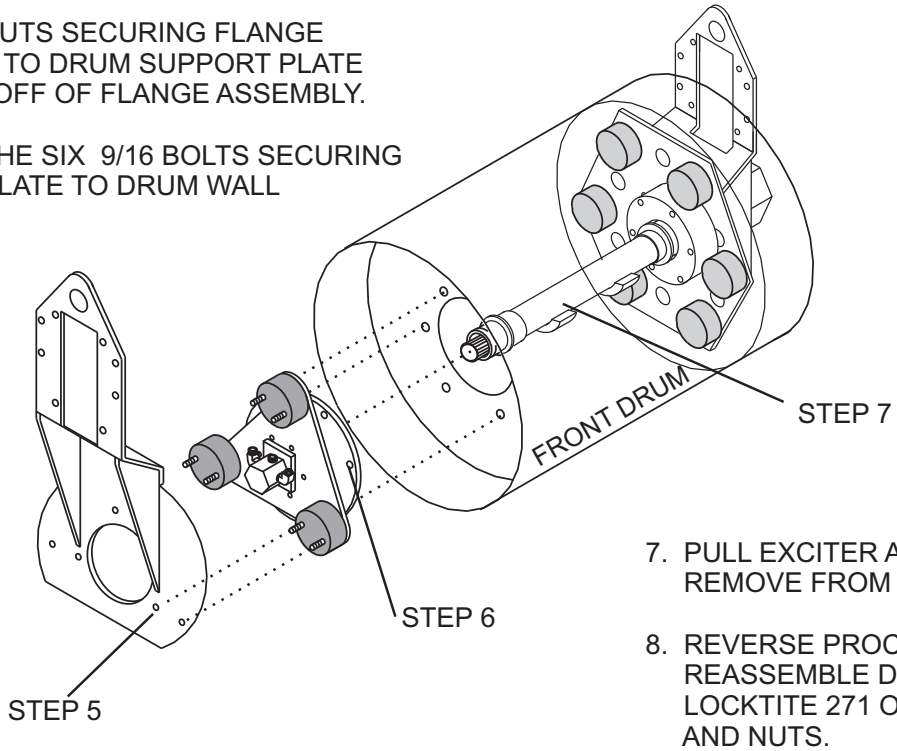


3. MARK AND DISCONNECT HOSES FROM MOTOR AND MANIFOLD.

4. REMOVE FOUR ALLEN HEAD SCREWS FROM DRUM SUPPORT PLATES.(SCREWS MAY NEED TO BE HEATED TO REMOVE)

5. REMOVE NUTS SECURING FLANGE ASSEMBLY TO DRUM SUPPORT PLATE AND PULL OFF OF FLANGE ASSEMBLY.

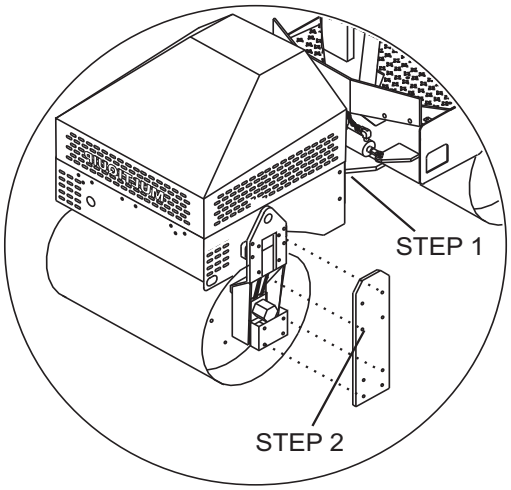
6. REMOVE THE SIX 9/16 BOLTS SECURING EXCITER PLATE TO DRUM WALL

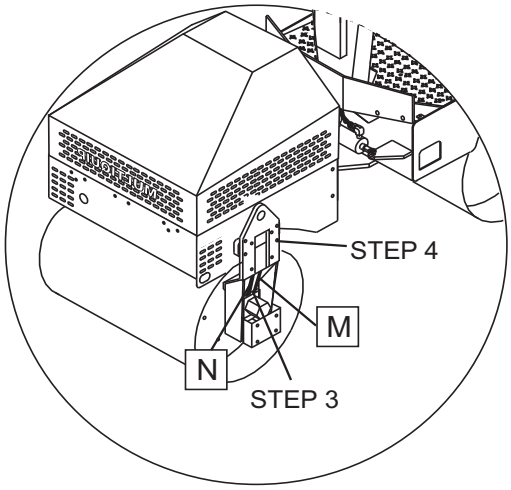


7. PULL EXCITER ASSEMBLY TO REMOVE FROM DRUM INTERIOR.

8. REVERSE PROCEDURES TO REASSEMBLE DRUM. USE LOCKTITE 271 ON ALL BOLTS AND NUTS.

Figure 37. Front-Right Drum Removal





1. JACK FRAME UP SLIGHTLY AT ARTICULATION JOINT.

2. REMOVE THE EIGHT ALLEN HEAD SCREWS HOLDING THE SIDE COVER OF THE DRUM SUPPORT PLATE.

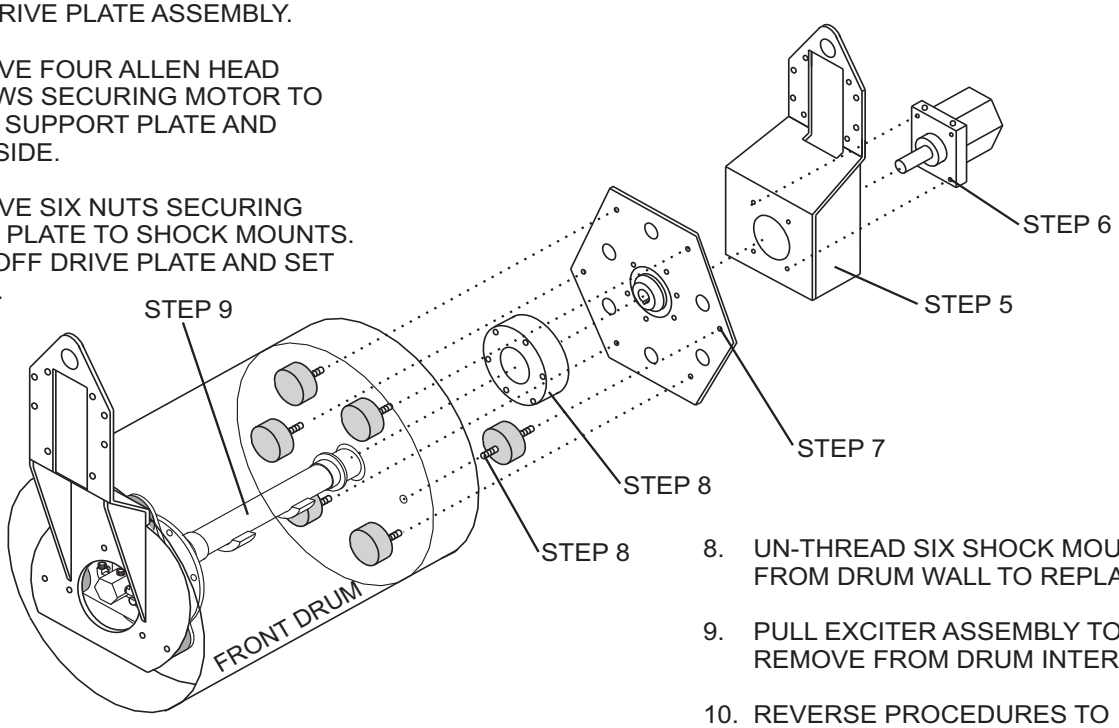
3. MARK AND DISCONNECT HOSES FROM MOTOR.

4. REMOVE FOUR ALLEN HEAD SCREWS FROM DRUM SUPPORT PLATE. (SCREWS MAY NEED TO BE HEATED TO REMOVE)

5. PULL DRUM SUPPORT PLATE OFF DRIVE PLATE ASSEMBLY.

6. REMOVE FOUR ALLEN HEAD SCREWS SECURING MOTOR TO DRUM SUPPORT PLATE AND SET ASIDE.

7. REMOVE SIX NUTS SECURING DRIVE PLATE TO SHOCK MOUNTS. PULL OFF DRIVE PLATE AND SET ASIDE.

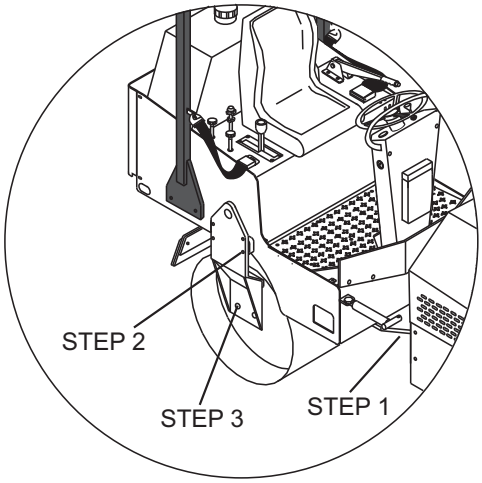


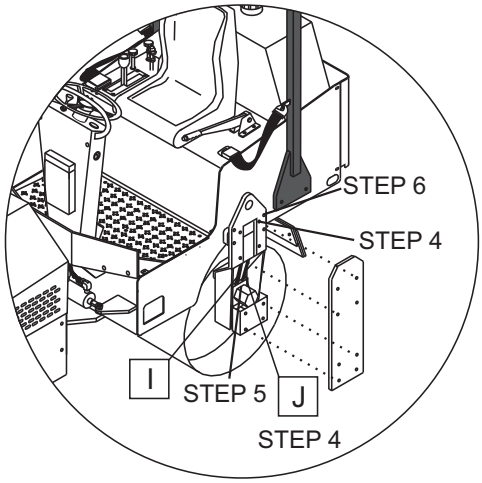
8. UN-THREAD SIX SHOCK MOUNTS FROM DRUM WALL TO REPLACE.

9. PULL EXCITER ASSEMBLY TO REMOVE FROM DRUM INTERIOR.

10. REVERSE PROCEDURES TO REASSEMBLE DRUM. USE LOCKTITE ON ALL NUTS AND BOLTS.

Figure 38. Front-Left Drum Removal





1. JACK FRAME UP SLIGHTLY AT ARTICULATION JOINT.

2. REMOVE THE FOUR ALLEN HEAD SCREWS SECURING THE DRUM SUPPORT PLATE TO THE FRAME.

3. REMOVE NUTS SECURING BEARING FLANGE TO BASE OF DRUM SUPPORT PLATE.

4. REMOVE ALLEN HEAD SCREWS HOLDING THE SIDE COVER OF THE DRUM SUPPORT PLATE.

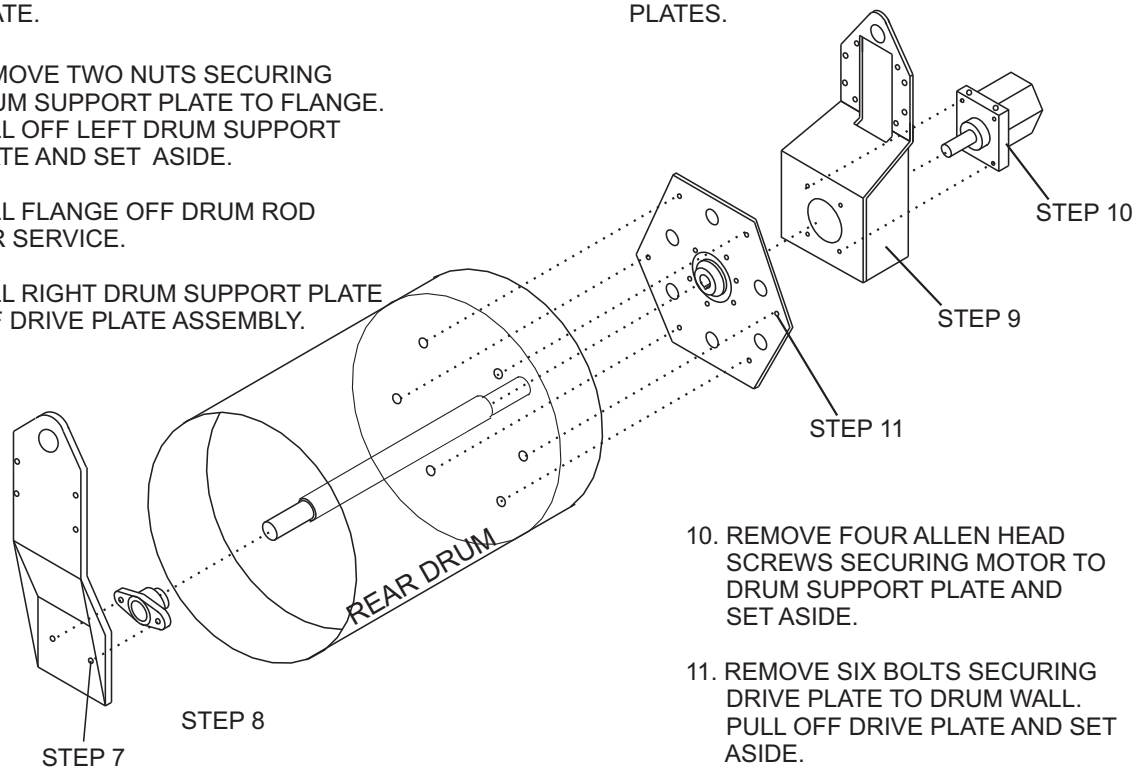
5. MARK AND DISCONNECT HOSES FROM MOTOR.

6. REMOVE FOUR ALLEN HEAD SCREWS FROM DRUM SUPPORT PLATES.

7. REMOVE TWO NUTS SECURING DRUM SUPPORT PLATE TO FLANGE. PULL OFF LEFT DRUM SUPPORT PLATE AND SET ASIDE.

8. PULL FLANGE OFF DRUM ROD FOR SERVICE.

9. PULL RIGHT DRUM SUPPORT PLATE OFF DRIVE PLATE ASSEMBLY.



10. REMOVE FOUR ALLEN HEAD SCREWS SECURING MOTOR TO DRUM SUPPORT PLATE AND SET ASIDE.

11. REMOVE SIX BOLTS SECURING DRIVE PLATE TO DRUM WALL. PULL OFF DRIVE PLATE AND SET ASIDE.

12. REVERSE PROCEDURES TO REASSEMBLE DRUM.

Figure 39. Rear Drum Removal/Installation

Alternator/Regulator

The AR-13H is equipped with a 20 amp charging system. This system uses three charge coils connected in parallel. The windings of each coil are wound with heave-duty insulated wire and are further protected by insulating material for long life. A voltage regulator is provided to control the amount of charge voltage being delivered to the 12 volt electrical system. For servicing this charging system, contact your Honda dealer.

Neutral Safety Switch

The ***travel lever*** is provided with a ***neutral safety switch*** (Figure 40) that prevents the engine from starting when the lever is in the forward or reverse drive position. Lever must be placed in the neutral position in order for the starter to function.

To check this switch, remove the panel located below and to the rear of the floor. This switch is mounted to the lower forward/reverse lever mechanism. Disconnect the ***white*** and ***green*** wires. Place the lever in neutral and test for continuity. Place the lever in forward or reverse; no continuity should be present. Repair or replace as required.

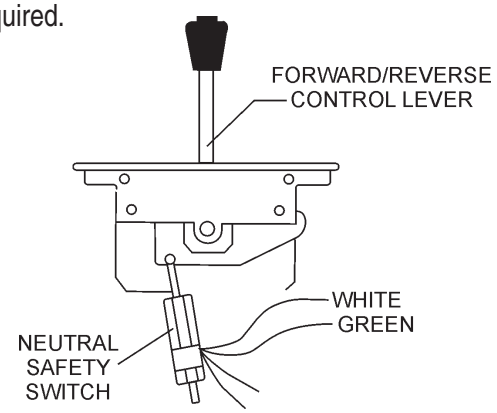


Figure 40. Safety Switch

Vibrator Solenoid

The vibrator solenoid (Figure 41) is located in the manifold valve block and is controlled by the vibration switch located on top of the travel lever. This solenoid controls the vibration control valve which supplies hydraulic oil to the vibration drive motor.

To test the vibration solenoid, turn the ignition switch on. Check for proper supply voltage and ground. If proper voltage and ground are present, check the solenoid for continuity. If no continuity, replace the solenoid coil. If continuity is present, place your hand on the solenoid and turn the switch to the **ON** position; movement should be felt. **DO NOT START THE ENGINE FOR THIS TEST**

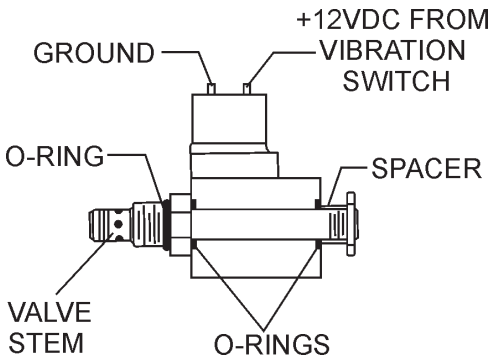


Figure 41. Vibration Solenoid

Battery

Mishandling of the battery shortens the service life of the battery ans adds to maintenance cost. When handling the battery do the following:

- Be careful not to let the battery electrolyte come in contact with your body or clothing.
- Always wear eye protection and rubber gloves, since the battery contains sulfuric acid which burns skin and eats through clothing.

⚠ CAUTION - BATTERY ACID BURN HAZARD

If your clothing or skin comes in contact with the battery acid, ***immediately*** wash off with running water and get medical attention.

- Check the battery regulary and make sure that each electrolyte level is to the bottom of the vent well (Figure 42). If necessary add only distilled water in a well-ventilated area.

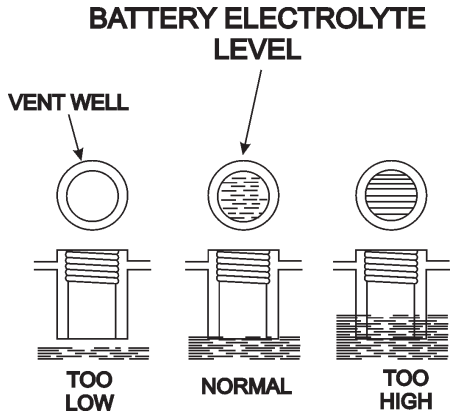


Figure 42. Battery Fluid Levels

RIDE-ON ROLLER — PREPARATION FOR LONG-TERM STORAGE

Roller Storage

For storage of the roller for over 30 days, the following is recommended:

- Drain the fuel tank completely, or add STA-BIL to the fuel.
- Run the engine until the fuel in the injection system is completely consumed**.**
- Completely drain used oil from the engine crankcase and fill with fresh clean oil, then follow the procedures described in the engine manual for engine storage.
- Drain water tank
- Clean the entire roller and engine compartment.
- Remove battery and store it in cool dry place.
- Cover the roller and place it a clean dry area, that is protected from harsh elements.
- Remove ignition key, and store in a safe place.

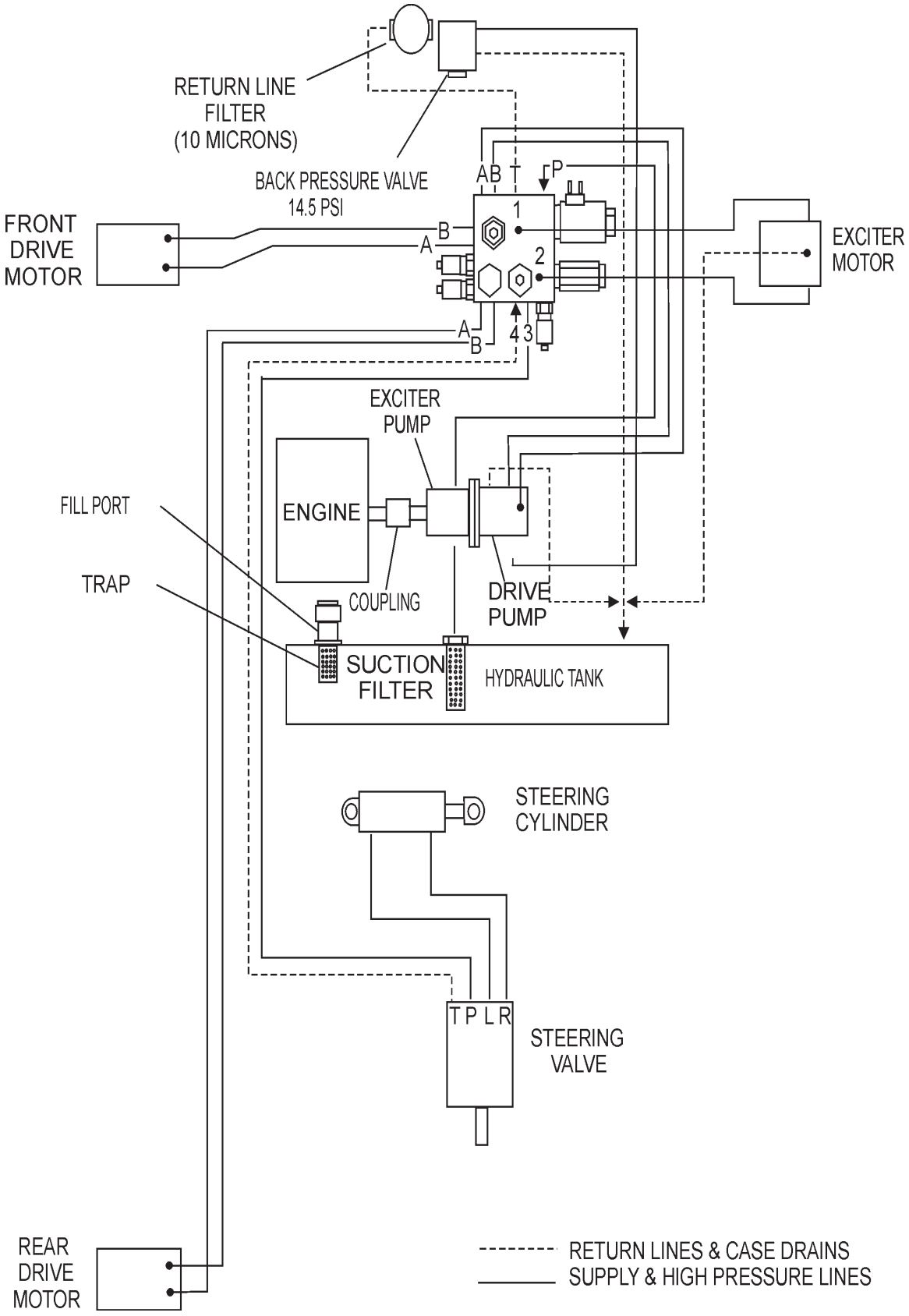


Figure 29. Manifold Test Ports

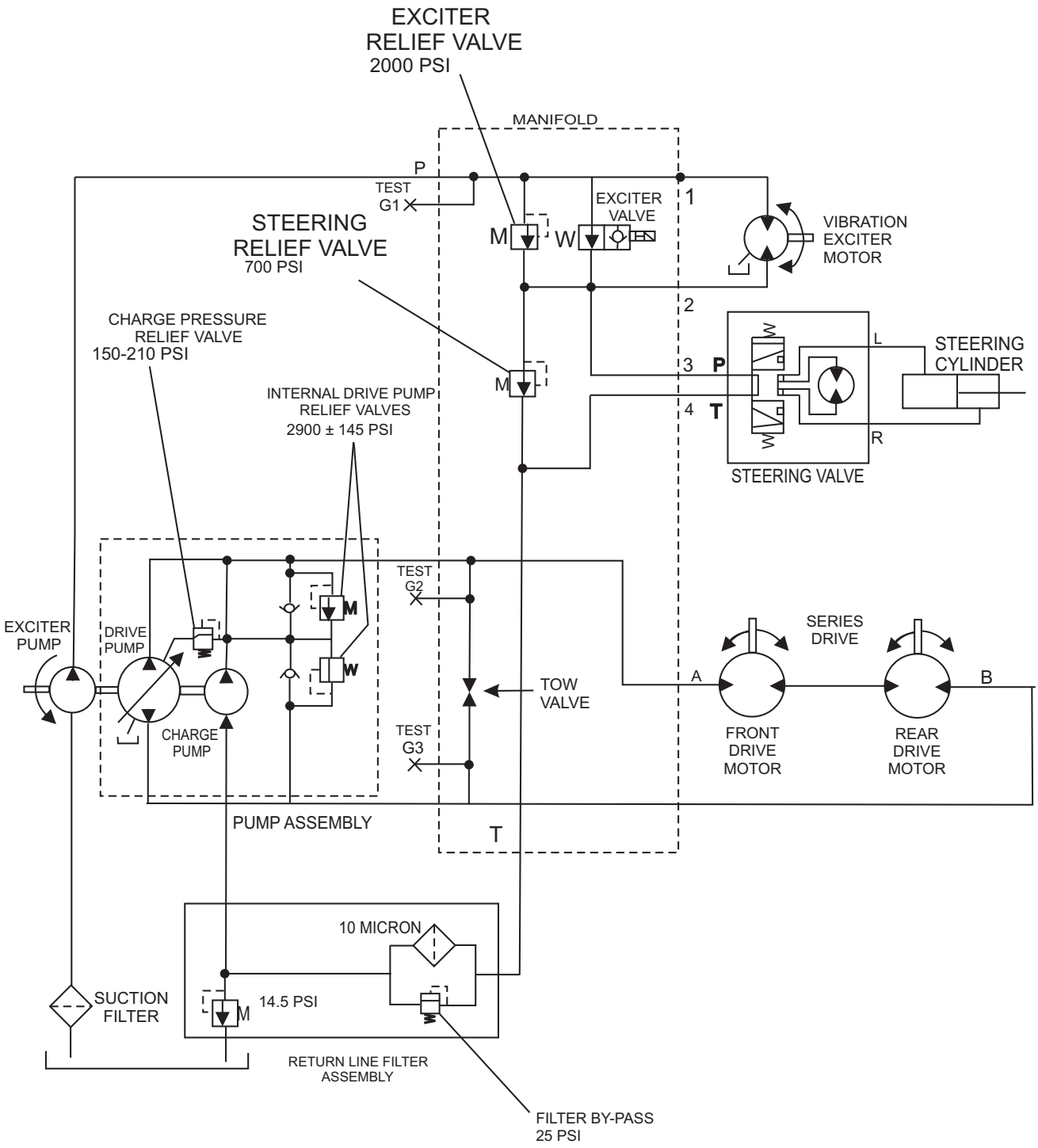
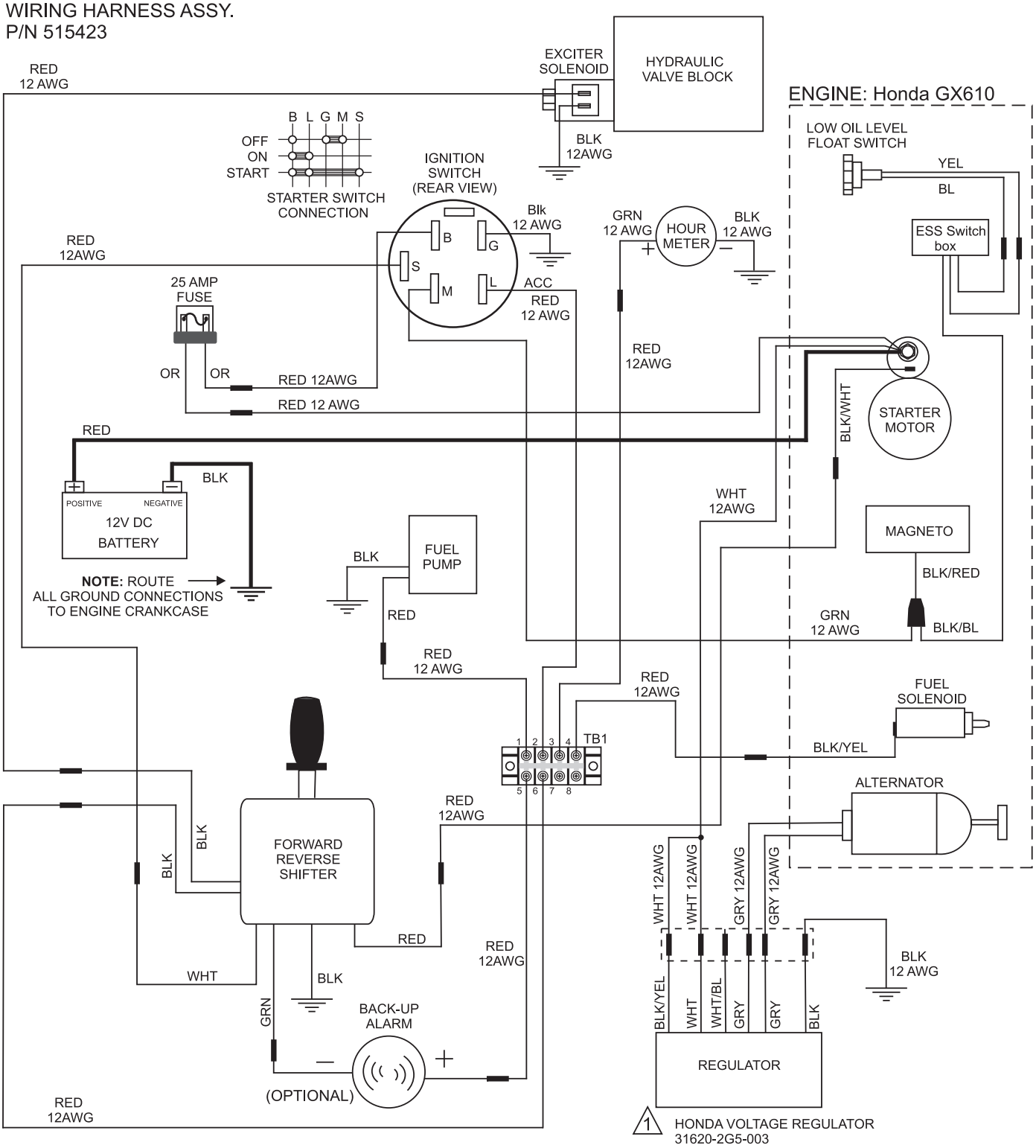


Figure 30. Hydraulic System Diagram

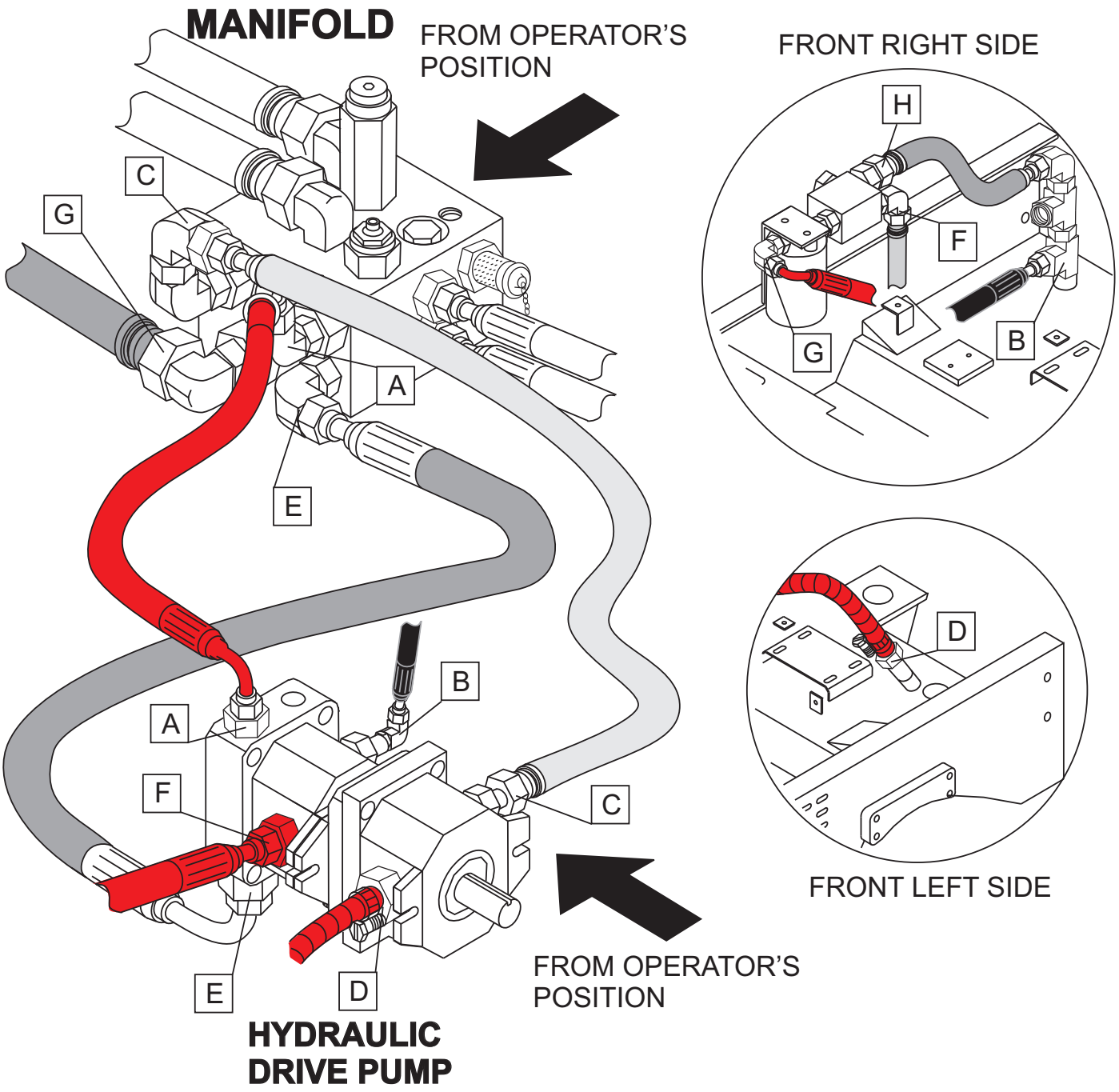


NOTE:

1 WHITE WIRE WITH BLUE STRIPE ON REGULATOR IS NOT USED.

BL = BLUE	BLK = BLACK
GRY = GRAY	YEL = YELLOW
GRN = GREEN	WHT = WHITE
OR = ORANGE	

NOTES



NOTES

HOSES SHADED FOR VISUAL CLARITY.

Figure 32. Hydraulic Hose Connections 1

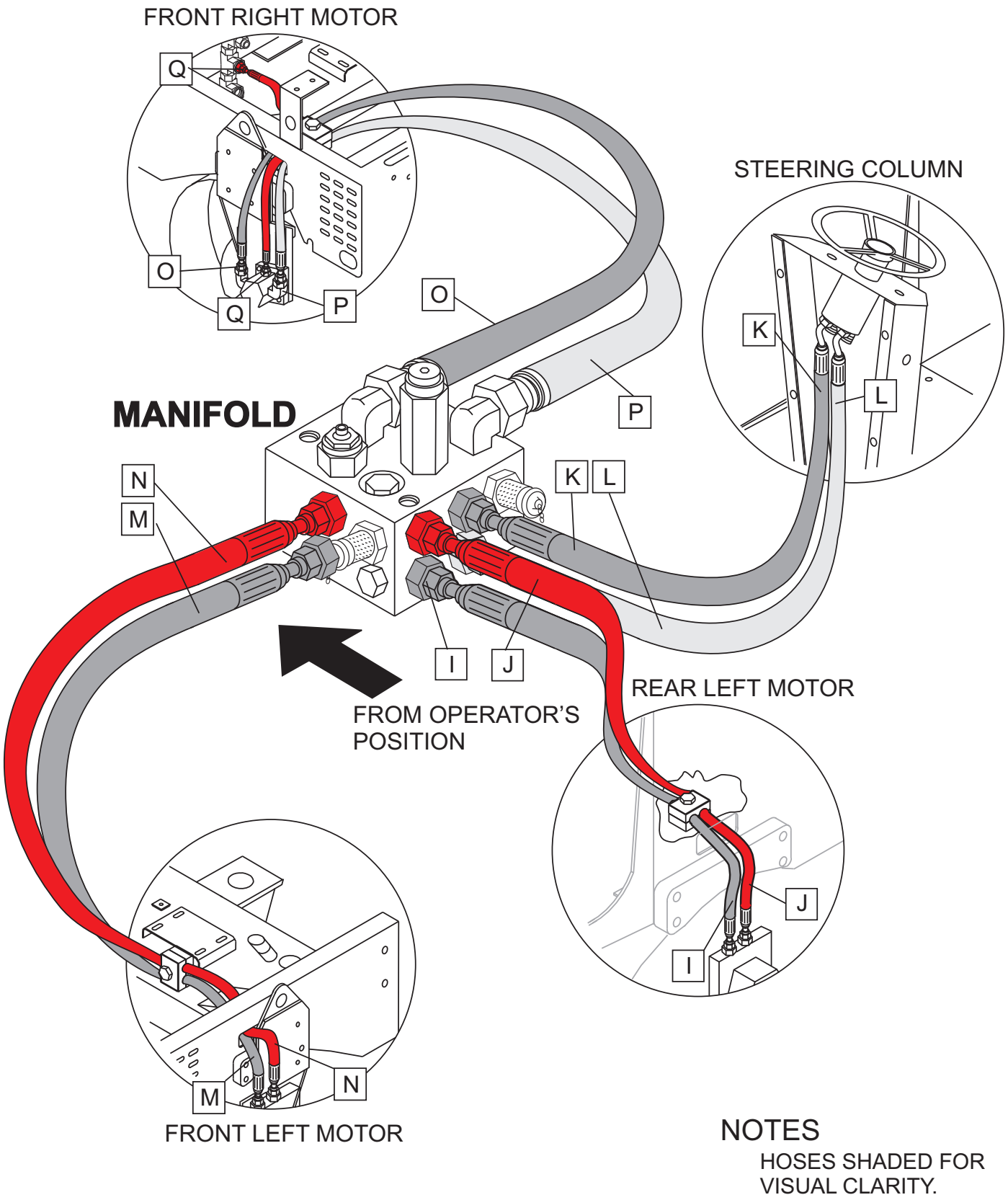


Figure 33. Hydraulic Hose Connections 2

Table 6. Roller Troubleshooting		
Symptom	**Possible Problem**	**Solution**
Low Vibration	Low engine RPM?	Check and adjust engine speed.
	Low hydraulic pressure?	Check and adjust hydraulic pressure, contaminant or defective relief valve.
	Vibrator shaft is binding	Remove motor from drum to determine if binding is result of shaft or motor.
No Vibration	No voltage to 12 volt solenoid?	Check vibration switch, check coil for continuity.
	No oil pressurel?	Check for proper oil pressure or defective relief valve.
	Broken motor coupler?	Replace coupler.
Slow Steering	Low hydraulic pressure?	Check pressure.
	Steering column connector?	Inspect bearings and grease if necessary.
	Defective steering cylinder?	Check seals, replace if necessary.
Steers Slow in One Direction	Defective Steering cylinder?	Inspect cylinder for contamination, replace if necessary.
Slow Forward/Reverse Speed	Low hydraulic pressure?	Check pressure.
	Rubber scrapper adjustment incorrect?	Check for proper adjustment against drum.
	Rubber shock mount loose?	Inspect for broken mount, replace if necessary.
	"Free wheel valve ""open""?"	Close valve.
	Speed control cable out of adjustment?	Inspect cable and adjust if necessary.
Main Hydraulic Pressure Low	Incorrect engine speed?	Adjust engine speed to correct speed.
	Broken rubber shock mounts or drive plate?	Inspect for cracks, replace if necessary.
	Drive motor hardware not secure?	Check motor nut and key coupler
	Relief valve	Check for contamination, replace.
	Drive motor	Replace motor.
	Broken motor shaft key or coupling	Replace motor shaft key or coupling.

TABLE 7. ENGINE TROUBLESHOOTING		
SYMPTON	**POSSIBLE PROBLEM**	**SOLUTION**
Engine will not start or start is delayed, although engine can be turned over.	No Fuel reaching injection pump?	Add fuel. Check entire fuel system.
	Defective fuel pump?	Replace fuel pump.
	Fuel filter clogged?	Replace fuel filter and clean tank.
	Faulty fuel supply line?	Replace or repair fuel line.
	Compression too low?	Check piston, cylinder and valves. Adjust or repair per engine repair manual.
	Fuel pump not working correctly?	Repair or replacefuel pump.
	Oil pressure too low?	Check engine oil pressure.
	Low starting temperature limit exceeded	Comply with cold starting instructions and proper oil viscosity.
	Defective battery?	Charge or replace battery.
At low temperatures engine will not start.	Engine oil too thick?	Refill engine crankcase with correct type of oil for winter environment.
	Defective battery?	Replace battery.
Engine fires but stops soon as starter is switched off.	Fuel filter blocked?	Replace fuel filter.
	Fuel supply blocked?	Check the entire fuel system.
	Defective fuel pump?	Replace fuel pump.
Engine stops by itself during normal operation.	Fuel tank empty?	Add fuel.
	Fuel filter blocked?	Replace fuel filter.
	Defective fuel pump?	Replace fuel pump.
	Mechanical oil pressure shutdown sensor stops the engine due to low oil?	Add oil. Replace low oil shutdown sensor if necessary.

TABLE 7. ENGINE TROUBLESHOOTING (CONTINUED)		
SYMPTON	**POSSIBLE PROBLEM**	**SOLUTION**
Low engine power, output and speed.	Fuel tank empty?	Replace fuel filter.
	Fuel filter clogged?	Replace fuel filter.
	Fuel tank venting is inadequate?	Ensure that tank is adequately vented.
	Leaks at pipe unions?	Check threaded pipe unions tape and tighten unions a required.
	Speed control lever does not remain in selected position?	See engine manual for corrective action.
	Engine oil level too full?	Correct engine oil level?
Low engine power output and low speed, black exhaust smoke.	Air filter blocked?	Clean or replace air filter.
	Incorrect valve clearances?	Adjust valves per engine specification.
	Malfunction at injector?	See engine manual.
Air-cooled diesel engine runs very "HOT". Cylinder head overheats.	Too much oil in engine crankcase?	Drain off engine oil down to uppermark on dipstick.
	Entire cooling air system contaminated blocked?	Clean cooling air system, and cooling fin areas.

NOTES

OPERATION MANUAL

HERE'S HOW TO GET HELP

PLEASE HAVE THE MODEL AND SERIAL NUMBER ON-HAND WHEN CALLING

UNITED STATES

Multiquip Inc.

(310) 537- 3700
6141 Katella Avenue Suite 200
Cypress, CA 90630
E-MAIL: mq@multiquip.com
WEBSITE: www.multiquip.com

CANADA

Multiquip

(450) 625-2244
4110 Industriel Boul.
Laval, Quebec, Canada H7L 6V3
E-MAIL: infocanada@multiquip.com

UNITED KINGDOM

Multiquip (UK) Limited Head Office

0161 339 2223
Unit 2, Northpoint Industrial Estate,
Globe Lane,
Dukinfield, Cheshire SK16 4UJ
E-MAIL: sales@multiquip.co.uk

© COPYRIGHT 2021, MULTIQUIP INC.

Multiquip Inc, the MQ logo and the Mayco logo are registered trademarks of Multiquip Inc. and may not be used, reproduced, or altered without written permission. All other trademarks are the property of their respective owners and used with permission.

This manual MUST accompany the equipment at all times. This manual is considered a permanent part of the equipment and should remain with the unit if resold.

The information and specifications included in this publication were in effect at the time of approval for printing. Illustrations, descriptions, references and technical data contained in this manual are for guidance only and may not be considered as binding. Multiquip Inc. reserves the right to discontinue or change specifications, design or the information published in this publication at any time without notice and without incurring any obligations.

MULTIQUIP®